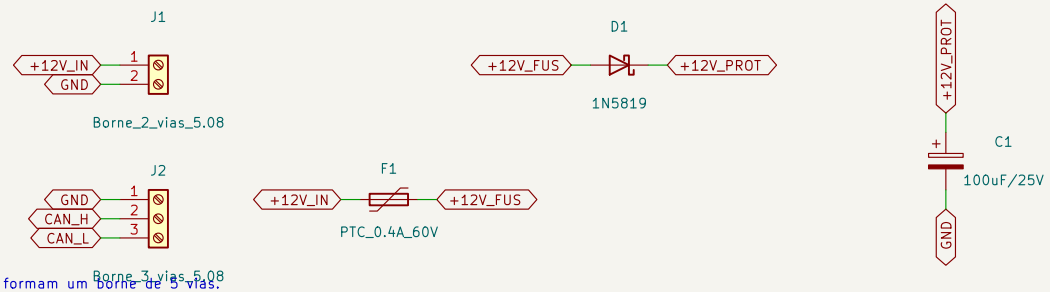
1. Entrada e protecao (cabo OBD-II)

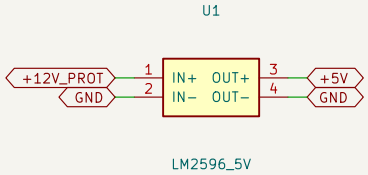


J1 + J2 formam um borne de 5 vias.

Cabo OBD-II: pino 16 = +12V, pinos 4 e 5 = GND, pino 6 = CAN_H, pino 14 = CAN_L

F1 (PTC rearmavel 0,4 A / 60 V): abre num curto na placa. D1 (1N5819): inversao de polaridade.

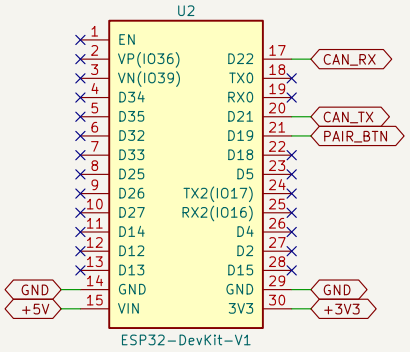
2. Regulacao 5 V (modulo LM2596)

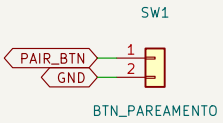


Ajustar a saida para 5,0 V ANTES de encaixar o ESP32.

Modulo encaixado em 4 soquetes femea de 1 pino.

3. Microcontrolador (DOIT ESP32 DevKit V1, 30 pinos)



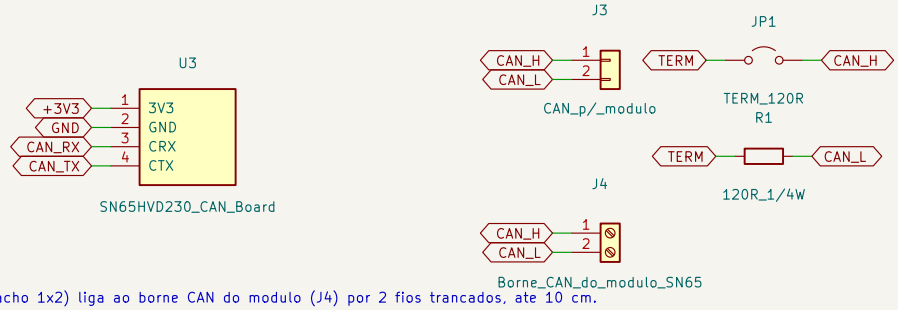


SW1: botao de pareamento (GPIO19 -> GND), externo. Segurar 5 s com o dongle ligado.

Alimentado pelo pino VIN (5 V). CAN: TX = GPIO21 (D21), RX = GPIO22 (D22).

Os demais pinos ficam sem ligacao (marcados com X).

4. Transceptor CAN (SN65HVD230 Waveshare, em pe)

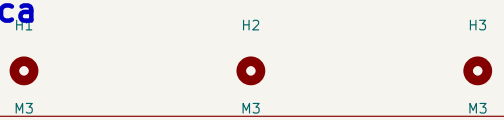


J3 (macho 1x2) liga ao borne CAN do modulo (J4) por 2 fios trancados, ate 10 cm.

JP1 colocado = terminacao de 120 ohms (bancada). Sem jumper = sem terminacao (carro).

J4 e so o borne do modulo (nao vai na PCB). Remover SEMPRE o resistor SMD de 120 ohms do modulo.

5. Mecanica



Alimentacao 12 V -> 5 V (LM2596) -> ESP32 + SN65HVD230 (CAN)

TCC ELTA

Sheet: /

File: tccelta.kicad_sch

Title: Dongle OBD-II - TCC

Size: A4	Date: 2026-11-07	**Rev: 1**
KiCad E.D.A. 10.0.8		Id: 1/1